

YUSEF AHMED KOURA

AI / Machine Learning Engineer & Mechatronics Engineering Graduate

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1. Professional Summary

Yousef Ahmed Koura is a Mechatronics Engineering graduate (GPA 3.62 / 4.0) from MSA University (supervised by University of Greenwich, UK) with hands-on experience building and deploying ML and deep learning models across computer vision, classification, and data pipeline workflows. He is a published researcher who deployed real-time AI on edge hardware (NVIDIA Jetson Nano) achieving 96% accuracy at 20 FPS. Skilled in Python, PyTorch, TensorFlow, and Docker, with practical exposure to MLOps and cloud-adjacent tools. Actively seeking roles in AI/ML engineering, deep learning, or data engineering.

2. Education

High School Diploma

Institution: Ibn Khaldoun School | Graduation Date: June 2019 | Grade: 97 / 100

B.Sc. Mechatronics Systems Engineering

Institution: MSA University (supervised by University of Greenwich, UK) | Location: Giza, Egypt

Start Date: September 2020 | End Date: June 2024 | Duration: 4 years | GPA: 3.62 / 4.0

Relevant Coursework: Stress Analysis, Computer Programming, Electric Drive System, Sensors, Mechatronics Modelling and Simulation, Project Management, CAD/CAM, Quality Management, Mechanical Vibration, Manufacturing Process, Robotics, Automatic Control, Microprocessor and Microcontroller, Artificial Intelligence, Hydraulic & Pneumatic Control, Industrial Automation.

3. Work Experience

Yousef has approximately 8 month of professional experience, spanning 4 roles: 2 AI/ML internships, 1 remote ML engineering role, and 1 engineering internship. Total hands-on ML/AI experience: approximately 7 months of internships plus ongoing project work.

Role 1: Machine Learning Engineer (Remote)

Organization: PioPetro | Location: Ohio, United States (Remote)

Start Date: June 2024 | End Date: August 2024 | Duration: 3 months

- Completed a virtual training program focused on ML applications in the petroleum industry.
- Covered domains: Petroleum Exploration, Artificial Lift Technology, Well Intervention, Well Stimulation.
- Applied supervised learning techniques to petroleum datasets as part of structured ML pipelines.

Role 2: AI/ML Intern

Organization: Information Technology Institute (ITI) | Location: Al Minufiyah, Egypt

Start Date: July 2023 | End Date: September 2023 | Duration: 3 months

- Developed and benchmarked supervised and unsupervised ML models (classification, regression, clustering) using Scikit-learn.
- Applied cross-validation and performance metrics (Accuracy, F1, RMSE) across structured datasets.
- Gained foundational experience in Python-based ML workflows and model evaluation.

Role 3: AI/ML Intern

Organization: Digital HUB (D-HUB) | Location: Cairo, Egypt

Start Date: August 2023 | End Date: August 2023 | Duration: 1 month

- Built a breast cancer detection classifier achieving 95% accuracy.
- Awarded 3rd place in a competitive AI challenge co-hosted with the Electronics Research Institute.

Role 4: Engineering Intern

Organization: Elmarakby Steel | Location: Egypt

Start Date: February 2023 | End Date: February 2023 | Duration: 1 month

- Completed a general training program across all engineering departments: Production, Maintenance, Quality, R&D, Hydraulic, Refractory, Electrical, and Safety.
- Assisted engineers in quality control checks, troubleshooting equipment issues, and analyzing data to improve production processes.

Total Work Experience Summary:

- Earliest role: February 2023 (Elmarakby Steel)
- Latest role: August 2024 (PioPetro)
- Total span: approximately 1.5 years (Feb 2023 – Aug 2024)
- Total months in ML/AI roles: approximately 7 months (ITI: 3 months + D-HUB: 1 month + PioPetro: 3 months)

4. Projects

Agri-Bot — AI-Powered Crop Disease Detection Robot

Type: Graduation Project | Start: November 2023 | End: June 2024 | Duration: 8 months

Designed a real-time computer vision pipeline using YOLOv8 (detection) and MobileNetV2 (classification). Deployed on NVIDIA Jetson Nano achieving 96% accuracy at 20 FPS. Implemented fuzzy PID motor control for autonomous navigation. Built a web interface for remote monitoring. Published as a research paper at IUGRC 2024. GitHub: <https://github.com/Yousef-koura/AGRI-BOT>

FMCG Data Engineering Pipeline

Type: Data Engineering | Year: 2026

Built an end-to-end Lakehouse pipeline on Databricks using PySpark, Delta Lake, and Unity Catalog. Implemented Medallion Architecture (Bronze → Silver → Gold) with Star Schema modeling. Configured incremental loading and orchestrated Databricks jobs. Created a live Genie AI dashboard. GitHub: https://github.com/Yousef-koura/FMCG_Data_Engineer_Project

Chest X-Ray Pneumonia Classification (PyTorch)

Type: Deep Learning / Medical Imaging | Year: 2026

Comparing custom CNN versus ResNet18 transfer learning on the Kaggle COVID-19 Radiography Dataset. Handles class imbalance using pos_weight in BCEWithLogitsLoss. Trained on GTX 1050 GPU with PyTorch 2.0.1 (CUDA 11.8).

Ball Tracking Mobile Robot

Type: Computer Vision / Robotics | Date: May 2023

Built a sensor-driven robot with real-time vision inference pipeline. Achieved 95% object detection accuracy and sub-5cm tracking error. GitHub: <https://github.com/Yousef-koura/Ball-Tracking-Robot>

Steganography Image Detection

Type: Deep Learning / Image Forensics | Date: February 2023

Trained a CNN-based classifier to detect hidden data in digital images. Achieved 90% classification accuracy. GitHub: <https://github.com/Yousef-koura/Steganography-Detector>

Breast Cancer AI Classifier

Type: Machine Learning / Classification | Date: August 2023

Built a classifier for early breast cancer detection achieving 95% accuracy. Awarded 3rd place at a competitive AI challenge co-hosted with the Electronics Research Institute. GitHub: <https://github.com/Yousef-koura/Breast-Cancer-Ai>

Machine Learning Projects Repository

Type: ML Portfolio | Period: 2023 – ongoing

9+ ML projects covering regression, classification, and clustering. Tools: Python, Pandas, Scikit-learn, Matplotlib, NumPy. GitHub: <https://github.com/Yousef-koura/Machine-Learning-Projects>

Deep Learning Projects Repository

Type: DL Portfolio | Period: 2023 – ongoing

Advanced neural network projects including CNNs, RNNs, and Transformers. Tools: TensorFlow, Keras, PyTorch. GitHub: https://github.com/Yousef-koura/Deep_Learning_Projects

SQL Projects Repository

Type: Data / SQL | Period: 2024 – ongoing

Data analysis and management using complex SQL queries. Covers database design and optimization in PostgreSQL and MySQL. GitHub: https://github.com/Yousef-koura/SQL_Projects

Vertical Axis Wind Turbine

Type: Hardware / Renewable Energy | Date: April 2020

Designed and built a vertical axis wind turbine. Conducted research, modeled using CAD software, tested prototype, and worked on control and electrical systems.

Human Following Robot

Type: Robotics / Embedded Systems | Date: April 2021

Designed a robot capable of following a human using sensors, microcontroller programming, and actuator integration.

Automatic Color Sorting Machine

Type: Automation / Embedded Systems | Date: June 2022

Built a machine with color sensors, servo motors, and sorting algorithms. Responsible for mechanical design, microcontroller programming, and sensor integration.

Hot Wire Cutting CNC Machine

Type: CNC / Manufacturing | Date: December 2022

Built a CNC machine using a heated wire to cut foam, Styrofoam, and plastic. Covered mechanical design, CNC programming, and component integration.

5. Skills

Programming Languages: Python, C++, C, MATLAB, SQL

ML / Deep Learning: Supervised Learning, Unsupervised Learning, Neural Networks (CNNs, RNNs), Transfer Learning, Object Detection (YOLOv8), LLMs, RAG, Classification, Regression, Clustering

Frameworks & Libraries: TensorFlow, PyTorch, Keras, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, FastAPI, OpenCV, LangChain

Data Engineering: ETL/ELT, Medallion Architecture, Delta Lake, PySpark, Databricks, Star Schema

MLOps & DevOps: Git, GitHub, Docker, Kubernetes, REST APIs, ONNX Runtime, AWS (fundamentals), CI/CD

Hardware & Robotics: NVIDIA Jetson Nano, Microcontrollers, ROS, PID/Fuzzy Control, Sensors & Actuators, Embedded Systems

Design & CAD: SolidWorks, AutoCAD

Soft Skills: Problem-solving, Time Management, Adaptability, Research, Leadership, Project Management, Technical Writing, Presentation

Languages: Arabic (Native), English (Professional)

6. Publications

Title: Agri-Bot: AI-Powered Crop Disease Detection and Medication Recommendation Robot

Published at: 8th International Undergraduate Research Conference (IUGRC 8) | Year: 2024

This paper presents an autonomous agricultural robot integrating YOLOv8 for real-time crop disease detection and MobileNetV2 for classification, deployed on NVIDIA Jetson Nano. The system achieves 96% accuracy at 20 FPS, uses fuzzy PID motor control for navigation, and includes a medication recommendation module and web interface.

PDF: https://github.com/Yousef-koura/yousefkoura.github.io/raw/main/AgRobot_paper.pdf

7. Certifications

- Machine Learning Specialization — Coursera | Year: 2023

- Supervised Machine Learning: Regression and Classification — Coursera | Year: 2023

- Python 101 for Data Science — IBM Cognitive Class | Year: 2022

- Certified in Learning Excel: Data Analysis — LinkedIn | Year: 2022
- Arduino Platform and C Programming — Coursera | Date: April 2022

Total Certifications: 5

8. Personal Information & Contact

Full Name: Yousef Ahmed Koura

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LinkedIn: linkedin.com/in/yousefkoura | linkedin.com/in/yousefahmedkoura

Portfolio: yousefkoura.github.io

Military Service: Completed

Availability: Available for work immediately

9. Quick Facts (Summary for Q&A)

- Full name: Yousef Ahmed Koura
- Degree: B.Sc. Mechatronics Systems Engineering from MSA University (2020–2024)
- GPA: 3.62 / 4.0
- Total number of work experience roles: 4
- Total ML/AI work experience: approximately 7 months
- Total professional experience span: (approximately 8 months)
- Total projects: 30 projects (hardware, ML, DL, data engineering)
- Total AI/ML projects: 30 (Agri-Bot, Breast Cancer Classifier, Ball Tracking Robot, Steganography Detector, FMCG Pipeline, Chest X-Ray, ML Repository, DL Repository, SQL Projects)
- Total certifications: 5
- Total publications: 1 (IUGRC 2024)
- Primary programming language: Python
- Current location: Menoufia, Egypt
- Job status: Actively seeking AI/ML or Data Engineering roles